

Hendrik Nicolas Carlo

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SUMMARY

Computer Science undergraduate (GPA 3.95) and Apple Developer Academy graduate specializing in Applied AI, Computer Vision, and ML deployment. Published IEEE researcher with hands-on experience shipping production iOS applications and scalable ML inference systems. Proven ability to bridge research-grade models and real-world mobile deployment using CoreML, Swift, Docker, and cloud infrastructure.

EDUCATION

- Bina Nusantara University, School of Computer Science** Jakarta, Indonesia
Bachelor of Technology - Information Technology; GPA: 3.95/4.00 Sept 2022 – Feb 2026
 - Relevant Coursework:** Analysis of Algorithms, Data Structures, Artificial Intelligence, Machine Learning, Computer Vision, Software Engineering, Operating Systems, Database Systems.

EXPERIENCE

- Apple Developer Academy @BINUS** Tangerang, Indonesia
Junior AI Engineer & iOS Developer (Internship) March 2025 – Dec 2025
 - Product Development & Deployment:** Developed **5 iOS applications** in 10 months, resulting in **2 App Store launches** and **2 TestFlight betas**. Applied the **Challenge Based Learning (CBL)** framework to drive iterative product design, validation, and deployment.
 - Selected Technical Projects:**
 - * **Rulaa** — Clothing Size Recommendation & Body Measurement App
Led the **end-to-end AI/ML development lifecycle**, including **model benchmarking and performance evaluation**, **ML infrastructure research and server architecture design**, and **Docker-based containerization** to enable **serverless deployment** of scalable inference services. Developed the iOS client using **SwiftUI**, **UIKit**, and **Vision** following the **MVVM** architecture with secure **API integration** to production ML endpoints.
 - * **Buah Hati** — Avocado Ripeness Classification App
Designed and implemented the **data engineering and ML pipeline**, including **dataset cleaning and preprocessing**, **automated object detection annotation**, and **on-device model deployment** using **CoreML**. Built the application with **SwiftUI**, **SwiftData**, and **Vision**, enabling **offline-first, low-latency inference** for end users.
 - Leadership & Research:** Founded and led the internal **Machine Learning Club**, delivering technical workshops on the **Apple AI ecosystem**. Prototyped **2 macOS applications** leveraging **CoreML**, **Vision**, and **CreateML** to demonstrate production-ready on-device intelligence systems.

PUBLICATIONS

- Comparative Analysis of Fungal Infections Classification in Apple Leaves**
Published in *6th International Conference on Cybernetics and Intelligent System (ICORIS), IEEE, 2024*
 - Co-authored a paper comparing standard CNNs vs. CNNs with texture analysis (GLCM), providing critical analysis on feature extraction efficiency.
 - Authors:** Carlo, H. N., Andreas, K., & Zakiyyah, A. Y.
- JKTDrive: Early Explorations of Visual-Language Model on Autonomous Driving in Indonesia**
Accepted in *12th International Conference on Computing and Artificial Intelligence (ICCAI), IEEE, 2026*
 - Co-authored a paper Benchmarking VLM performances as a reasoning model for Autonomous Driving Agent in Indonesia.
 - Authors:** Carlo, H. N., Sanjaya, J. S., Tanjaya V., & Cenggoro, T. W.

ACHIEVEMENTS

- ICPC 2023 (Top 60 Binus) - Nov 2023
- Top 20 Microsoft AI for Accessibility - May 2024
- Semi-finalist Samsung Solve for Tomorrow - Aug 2025

SKILLS SUMMARY

- Languages:** Python, Swift, C++, Bash, SQL, JavaScript, Java
- AI & Data:** TensorFlow, PyTorch, Keras, Scikit-Learn, Pandas, NumPy, OpenCV, CoreML
- Backend & Tools:** FastAPI, RESTful API, Docker, Git, AWS, GCP
- Soft Skills:** Technical Writing, Research, Public Speaking, Agile Leadership